Retrieval-Augmented Simulacra: Generative Agents for Up-to-date and Knowledge-Adaptive Simulations

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***Abstract*—Social networking services (SNS) are widely used across national boundaries, and extensive research has been conducted on their applications in marketing as well as the dissemination of emotions and information through such platforms. However, predicting user behavior on SNS remains a significant challenge, underscoring the need for systems capable of modeling and forecasting interaction tendencies. To address this, we have developed an SNS simulation system composed of large language model (LLM)-driven agents. This study investigates the impact of a retrieval-augmented generation mechanism, designed to emulate human-like search behavior, on the quality of generated posts and replies. The experimental results indicate that the proposed mechanism produces the most natural and coherent exchanges among agents.**

***Index Terms*—Large Language Model, Social Networking, Multi agent simulation, Retrieval Augmented Generation**

I. Introduction

Social networking services (SNS) have become an integral part of daily communication for many individuals. Furthermore, numerous companies utilize SNSs for marketing and market research, reflecting their widespread societal use.

A notable characteristic of SNS is the difficulty in predicting trends, which complicates the forecasting of phenomena such as "flaming" and the dissemination of misinformation. Consequently, there is a growing need for systems capable of predicting user behavior on SNS to ensure their safe and effective use. To address this, we propose a simulation system that models user interactions by employing multiple chat agents powered by large language models (LLM), enabling the emulation and prediction of SNS interactions.

II. Related Work

Ohagi [1] developed a discussion simulation system utilizing multiple agents driven by LLMs to investigate the effects of the echo chamber phenomenon on agent opinions. The study demonstrated that the echo chamber effect emerges in LLM-based agents when the system is configured to encourage more interactions among agents sharing similar views. These findings suggest that LLMs possess the capability to replicate and emulate human communication patterns and related social phenomena.

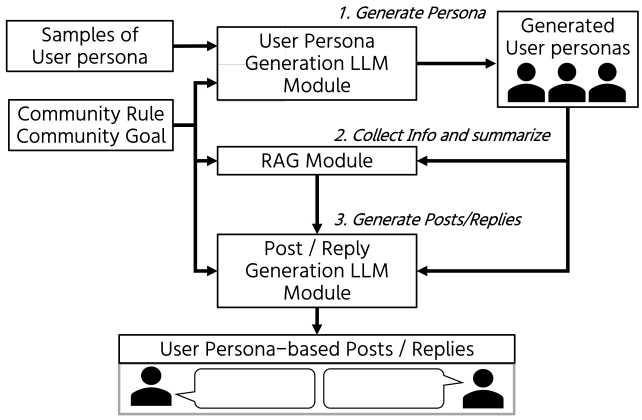

Fig. 1. Ovewview of system. It consists of two module, LLM module and RAG module that provides information to model.

Park et al. [2] conducted simulations of interactions on social networking sites using chat agents powered by LLMs. Building on their work, our study employs Retrieval-Augmented Generation (RAG) to enable simulations on up-to-date topics. Furthermore, we extended our evaluation to include sequences of posts and replies to better assess the system's ability to simulate interaction, which is a central aspect of social networking services.

In our previous work, we developed a social networking simulation system using a LLM with a RAG mechanism [3]. This study focuses on evaluating how the RAG mechanism improves the realism of the simulation.

III. Proposed Method

A. Overview of The Proposed Method

An overview of the system is shown in Figure 1. The simulation is conducted in three stages: user persona gen-

TABLE I

THE SUMMARIZATION TARGET AND THE TARGET LENGTH OF EACH SUMMARY. THE PERCENTAGES INDICATE THE RATIO OF THE NUMBER OF SENTENCES AFTER SUMMARIZATION TO THE NUMBER BEFORE SUMMARIZATION. THE "TARGET" COLUMN DEFINES THE SCOPE OF SUMMARIZATION: "TITLE" REFERS TO THE ARTICLE'S HEADLINE ONLY, "ABSTRACT" INCLUDES BOTH THE TITLE AND THE ABSTRACT, AND "MAIN BODY" COMPRISES THE TITLE, ABSTRACT, AND THE MAIN TEXT OF THE ARTICLE.

<table>
<tr><th rowspan="2">value of "depth"</th><th rowspan="2">Target</th><th colspan="2">Amount of indivisual sentences</th><th rowspan="2">Amount of total sentences</th></tr>
<tr><th>$l < 500$</th><th>$l \geq 500$</th></tr>
<tr><td>0</td><td>Title</td><td>100%</td><td colspan="2">100%</td></tr>
<tr><td>1</td><td>Abstruct</td><td>100%</td><td colspan="2">100%</td></tr>
<tr><td>2</td><td>Main body</td><td>10%</td><td colspan="2">5%</td></tr>
<tr><td>3</td><td>Main body</td><td>20%</td><td colspan="2">10%</td></tr>
<tr><td>4</td><td>Main body</td><td>30%</td><td colspan="2">15%</td></tr>
<tr><td>5</td><td>Main body</td><td>40%</td><td colspan="2">20%</td></tr>
<tr><td>6</td><td>Main body</td><td>50%</td><td colspan="2">25%</td></tr>
</table>

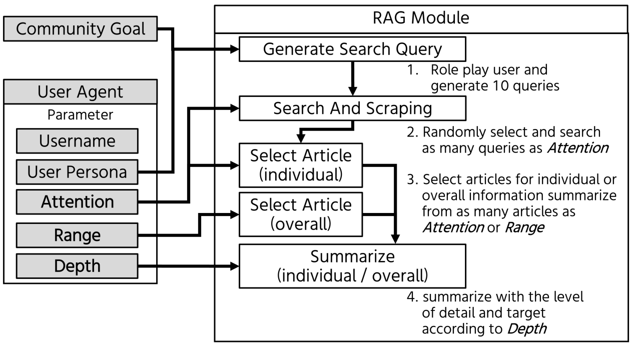

Fig. 2. RAG module. Based on the parameters given to the user persona, the number of search queries generated, the number of scraping articles, and the length of articles after summarization are determined.

eration, information collection & summarize, and post/reply generation.

At the beginning of each simulation run, the simulation operator provides three inputs: a community goal, a set of community rules, and sample user personas. The community goal defines the overall topic or purpose of the simulated discussion. The community rules specify the constraints and prohibitions that participants must adhere to. The sample user personas represent the expected user demographics within the community and serve as references for generating individual user personas used in the simulation.

B. Stage 1: Generate Personas

In this study, we define a "user persona" as a short textual description that characterizes a specific aspect of a user. For example, user personas may take the form of phrases such as "a university student studying social interaction" or "a fan of the NBA league". In generating user personas, we sample up to 10 user personas and create Few-shot prompts.

When generating user personas, three parameters: "attention", "range", and "depth" are assigned. The values of attention and range are randomly sampled from a normal distribution $\mathcal{N}(5, 1)$, rounded to the nearest integer, and clipped to lie within the range of 0 to 10. The depth parameter is sampled from an Erlang distribution $Er(3, 1)$, also rounded to an integer and constrained to the range of 0 to 6. Further details regarding these parameters are provided in Section III-C.

C. Stage 2: Information Collection & Summarize

To facilitate the simulation of discussions on up-to-date or domain-specific topics, our system incorporates Retrieval-Augmented Generation (RAG) [4] by leveraging information retrieved from the Web. As illustrated in Figure 2, relevant content is gathered and synthesized by multiple modules based on parameters assigned to each agent. This process emulates human information-seeking behavior, wherein information is collected in a multifaceted and context-sensitive manner.

First, the module generates ten search queries based on the community goal, user personas, and thread content. Then, a number of queries equal to the user's "attention" value are randomly selected, and for each selected query, the top ten articles from the search results are scraped. In this study, NHK NEWS WEB [1] was used as the source for web scraping.

The scraped articles are summarized using LexRank [5]. Both the part of article to be summarized and the target length of each summary are determined according to the criteria shown in Table I . A number of articles equal to the "range" and "attention" values are used to capture general and individual-level information, respectively. This approach emulates human information-gathering behavior, in which information is collected in a multifaceted and context-sensitive manner.

The incorporation of the RAG mechanism enables the simulation system to handle a wider range of topics, including up-to-date and domain-specific content. By providing agents with access to external information, the system can generate more realistic and contextually relevant interactions. This enhances the fidelity of the simulation, allowing for more accurate modeling of social phenomena such as public opinion formation or community response to current events.

D. Stage 3: Generate Posts & Replies

Posts and replies are generated based on prompts that incorporate user personas, community goals and rules, as well as the collected and summarized information. A certain

[1] https://www3.nhk.or.jp/news/

TABLE II

EVALUATION RESULTS OF WHETHER POSTS AND REPLIES ARE CONSISTENT WITH THE COMMUNITY GOAL UNDER THE "BANK OF JAPAN DISCUSSION" SCENARIO. THE ADV. RAG CONDITION ACHIEVED THE HIGHEST AVERAGE SCORE; HOWEVER, THE DIFFERENCES AMONG ALL CONDITIONS OR MODELS WERE GENERALLY SMALL.

Condition	Model	Consistent (score=5)	Somewhat consistent (score=4)	Somewhat inconsistent (score=2)	Inconsistent (score=1)	Total #	Ave. Score
Adv. RAG	Local	286	96	27	5	414	4.52
	GPT	345	59	8	2	414	**4.78**
	Gemini	318	88	8	0	414	4.73
Sim. RAG	Local	275	90	37	12	414	4.40
	GPT	326	73	9	6	414	**4.70**
	Gemini	301	90	20	3	414	4.61
w/o. RAG	Local	336	69	7	2	414	**4.76**
	GPT	334	70	7	3	414	4.75
	Gemini	279	107	25	3	414	4.53

TABLE III

EVALUATION RESULTS OF WHETHER THREADS ARE CONSISTENT WITH THE COMMUNITY GOAL UNDER THE "BANK OF JAPAN DISCUSSION" SCENARIO. GPT-4O-MINI ACHIEVED THE HIGHEST AVERAGE SCORE; HOWEVER, THE DIFFERENCES BETWEEN CONDITION OR MODELS WERE GENERALLY SMALL.

Condition	Model	Consistent (score=5)	Somewhat consistent (score=4)	Somewhat inconsistent (score=2)	Inconsistent (score=1)	Total #	Ave. Score
Adv. RAG	Local	56	20	5	0	81	4.57
	GPT	56	22	1	2	81	**4.59**
	Gemini	53	28	3	0	84	4.56
Sim. RAG	Local	46	23	7	2	78	4.33
	GPT	52	21	5	0	78	**4.54**
	Gemini	51	22	7	1	81	4.42
w/o. RAG	Local	52	16	6	1	75	4.49
	GPT	56	26	2	0	84	**4.62**
	Gemini	49	29	5	1	84	4.43

number of user personas are randomly selected to create top-level posts. A top-level post refers to an initial post that does not reply to any other post and serves as the starting point of a discussion thread. After that, all user personas generate replies to randomly selected posts and replies, simulating interactions within the community.

IV. EXPERIMENT

In this study, we evaluate posts, replies, and entire threads to examine the impact of the RAG mechanism integrated into the simulation system. For this purpose, we used three language models: rinna-llama-3-youko-70b-instruct-Q4_K_M, GPT-4o-mini, and Gemini 2.0-flash, along with a predefined set of user personas.

A. The Procedure of Evaluating Posts and Threads

In order to examine the impact of the RAG mechanism on the behavior of simulations, we conducted experiments under three different conditions.

- With the advanced RAG mechanism (proposed method)
- With simple RAG mechanism
- Without RAG mechanism

In "simple RAG mechanism" setting, the same set of retrieved and summarized information is uniformly provided to all agents, regardless of their individual user personas or generated posts. This condition serves as a baseline for comparison, where the retrieval process does not reflect individual differences among agents.

Posts (including replies) and threads were evaluated based on five criteria using a four-point Likert scale. For thread evaluation, only threads with at least one reply were considered. Evaluators were explicitly instructed to refer to "comparative posts" and "comparative threads" prepared by the authors as benchmarks representing the highest and lowest ratings for each category, such as most and least natural, and to base their ratings on these references. The evaluation criteria are as follows:

1) Are there any grammatical errors in the content of the post or thread?
2) Is the content of the post or thread appropriate for the intended purpose of the community interaction?
3) Does the content of the post or thread comply with the community rules?
4) Does the content of the post reflect the characteristics expected from the persona used to generate it? (evaluated only for posts/replies)
5) Is the content natural as a post on an online forum that shares the specified community goal?

Two simulation scenarios were employed in the experiments: "Bank of Japan Discussion" (a discussion on the Bank of Japan's monetary policy) and "Ohtani Chat" (a casual conversation about Shohei Ohtani, a professional baseball player known for his two-way role as both a pitcher and a hitter in Major League Baseball). These scenarios were chosen to represent topics that require either up-to-date or domain-specific information, thereby providing a suitable context for evaluating the effectiveness of the RAG mechanism.

TABLE IV

RESULTS OF EVALUATING WHETHER POSTS AND REPLIES ARE NATURAL IN THE COMMUNITY UNDER THE "BANK OF JAPAN DISCUSSION" SCENARIO. THE HIGHEST AVERAGE SCORE IN EACH CONDITION IS SHOWN IN BOLD. RESULTS INDICATE THAT THERE IS NO SIGNIFICANT DIFFERENCE BETWEEN THE CONDITIONS. NOTE THAT THIS IS THE RESULT OF AN EVALUATION OF A SINGLE POST AND REPLY.

Condition	Model	Natural (score=5)	Somewhat natural (score=4)	Somewhat unnatural (score=2)	Unnatural (score=1)	Total #	Ave. Score
Adv. RAG	Local	214	158	28	14	414	4.28
	GPT	284	112	13	5	414	4.59
	Gemini	290	108	13	3	414	**4.62**
Sim. RAG	Local	221	122	56	15	414	4.15
	GPT	292	104	12	6	414	**4.60**
	Gemini	253	135	22	4	414	4.48
w/o. RAG	Local	249	142	16	7	414	4.47
	GPT	288	113	12	1	414	**4.63**
	Gemini	252	137	21	4	414	4.48

TABLE V

RESULTS OF EVALUATING WHETHER POSTS AND REPLIES ARE NATURAL IN THE COMMUNITY UNDER THE "OHTANI CHAT" SCENARIO. THE HIGHEST AVERAGE SCORE IN EACH CONDITION IS SHOWN IN BOLD. AS WITH THE "BANK OF JAPAN DISCUSSION" SCENARIO, THE RESULTS INDICATE THAT THERE IS NO SIGNIFICANT DIFFERENCE BETWEEN THE CONDITIONS. NOTE THAT THIS IS THE RESULT OF AN EVALUATION OF A SINGLE POST AND REPLY.

Condition	Model	Natural (score=5)	Somewhat natural (score=4)	Somewhat unnatural (score=2)	Unnatural (score=1)	Total #	Ave. Score
Adv. RAG	Local	224	123	52	15	414	4.18
	GPT	283	111	12	8	414	**4.57**
	Gemini	229	144	29	12	414	4.33
Sim. RAG	Local	265	103	33	13	414	4.39
	GPT	287	109	13	5	414	**4.59**
	Gemini	231	140	37	6	414	4.34
w/o. RAG	Local	252	126	29	7	414	4.42
	GPT	281	109	20	4	414	**4.55**
	Gemini	241	142	27	4	414	4.42

B. Results of Evaluating Posts and Threads

Table II presents the evaluation results regarding whether individual posts and replies adhere to the community goal. Table III shows the corresponding results for entire threads, assessing whether the overall conversation aligns with the community goal. Across all conditions, scenarios, and models, no substantial differences were observed. This was consistent across multiple evaluation criteria, including adherence to community rules, presence of textual errors, and alignment with user personas. Due to space limitations, the detailed results for these items are omitted.

Table IV and Table V summarize the scores for each community scenario. In the "Bank of Japan Discussion" scenario, the w/o RAG condition yielded the highest average score, whereas in the "Ohtani Chat" scenario, the Sim. RAG condition outperformed the others. These results indicate that the superiority of the proposed method is not conclusive. Nevertheless, all conditions achieved relatively high scores, suggesting that the differences among them were generally small.

Table VI and Table VII present the evaluation results regarding the naturalness of threads within each community context. In both scenarios, the Adv. RAG condition, which represents the proposed method, achieved the highest average scores. The differences in average scores were more evident than those observed in the evaluation of individual posts and replies. Additionally, GPT-4o-mini attained the highest average score in the "Bank of Japan Discussion" scenario, while Gemini 2.0-flash achieved the highest score in the "Ohtani Chat" scenario. These findings suggest that the performance of each model may vary depending on the nature of the discussion topic.

V. DISCUSSION

The Adv. RAG method achieved the highest scores in thread-level naturalness across both scenarios, demonstrating its effectiveness in generating coherent interactions. In contrast, this advantage was not consistently observed at the level of individual posts and replies. These findings imply that variability in agents' information-gathering contributes to more natural dialogue flow, even if single utterance quality remains comparable.

A notable finding is that the Sim. RAG condition, in which all agents receive identical information, consistently yielded lower thread-level naturalness scores. This uniformity likely limited the diversity of perspectives, resulting in less realistic interactions. Since real-world users rarely share the exact same knowledge, the lack of informational variation may have undermined the authenticity of the simulated conversations. These results underscore the importance of modeling individual differences in information access to enhance realism in multi-agent simulations.

TABLE VI
RESULTS OF EVALUATING WHETHER THREADS ARE NATURAL IN THE COMMUNITY UNDER THE "BANK OF JAPAN DISCUSSION" SCENARIO. THE HIGHEST AVERAGE SCORE IN EACH CONDITION IS SHOWN IN BOLD. THE HIGHEST SCORES ARE OBTAINED IN THE ADV.RAG CONDITION.

Condition	Model	Natural (score=5)	Somewhat natural (score=4)	Somewhat unnatural (score=2)	Unnatural (score=1)	Total #	Ave. Score
Adv. RAG	Local	33	33	14	1	81	4.02
	GPT	36	27	12	6	81	3.93
	Gemini	39	31	12	2	84	**4.11**
Sim. RAG	Local	25	36	11	6	78	3.81
	GPT	35	30	8	5	78	**4.05**
	Gemini	38	24	17	2	81	3.98
w/o. RAG	Local	25	32	13	5	75	3.79
	GPT	35	33	13	3	84	**4.00**
	Gemini	34	33	14	3	84	3.96

TABLE VII
RESULTS OF EVALUATING WHETHER THREADS ARE NATURAL IN THE COMMUNITY UNDER THE "OHTANI CHAT" SCENARIO. THE HIGHEST AVERAGE SCORE IN EACH CONDITION IS SHOWN IN BOLD. THE HIGHEST SCORES ARE OBTAINED IN THE ADV.RAG CONDITION.

Condition	Model	Natural (score=5)	Somewhat natural (score=4)	Somewhat unnatural (score=2)	Unnatural (score=1)	Total #	Ave. Score
Adv. RAG	Local	38	35	9	2	84	4.17
	GPT	51	21	12	0	84	**4.32**
	Gemini	31	37	13	0	81	4.06
Sim. RAG	Local	31	27	19	4	81	3.77
	GPT	37	31	12	4	84	4.01
	Gemini	39	35	8	2	84	**4.20**
w/o. RAG	Local	29	38	11	3	81	3.98
	GPT	31	31	12	4	78	3.94
	Gemini	42	30	8	1	81	**4.28**

This study used only NHK NEWS WEB as the information source, which may have introduced topic-specific bias or limitations, particularly for domains such as celebrity gossip. To improve the realism and accuracy of the simulation, incorporating multiple sources will be necessary. Addressing this remains an important direction for future work.

VI. CONCLUSION

This study proposes a simulation system for social networking interactions, in which a LLM equipped with a RAG mechanism generates user posts based on a given community topic, a set of rules, and user-specific characteristics. The system is designed to emulate communication within a large-scale online community by modeling diverse user behaviors.

We focused on evaluating the impact of incorporating a RAG mechanism into the simulation process. The experimental results showed that the proposed method, which accounts for individual differences in information gathering and reading comprehension, produced the most natural interactions among the tested conditions. These results indicate that integrating user-specific retrieval processes improves the realism of simulated social interactions.

VII. LIMITATION

First, the evaluation of posts, replies, and threads was based on subjective judgment without objective or quantitative metrics. Despite instructions to reference comparative threads, potential bias remains, limiting reproducibility and generalizability.

Second, only NHK NEWS WEB was used as the information source. Other online media and SNS data were not included, which may have led to scenario-dependent score variations. A more comprehensive evaluation would benefit from incorporating diverse data sources to better reflect a wider range of scenarios.

Third, while simulating social interactions, this study does not address ethical concerns such as content bias, limiting assessment of the simulation's realism and reliability.

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